

Democracy Bench: Can a Deployed Language Model Be Governed as a Public Policy Delegate?

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Abstract

Language models are being placed in public-sector decision loops, where a natural hope is that they can act as *policy delegates*: reflect a real public’s preferences, be steered toward them at runtime, and follow those preferences as they change—while never trading away rights protections for popularity. We introduce **Democracy Bench**, an evaluation harness that operationalizes these four requirements against *human survey distributions* (British Social Attitudes microdata, 2022–2024) rather than an LLM judge or an abstract left–right axis. Its central design decision is a *legitimacy boundary* between *contestable* items (tax/spend, welfare, the NHS—where a delegate *should* track the public) and *rights-floor* probes (due process, appeal, equality, speech—where it must *hold* even against a majority). Using the harness we evaluate a five-rung “steering ladder”—context, output logit-bias, activation steering, weight fine-tuning, and prompt scaffolds—asking of each rung whether it can carry per-item public preference while holding rights floors. The answer is negative for every rung *except evidence-in-context*, which does let a model track a real 2022→2024 public shift (direction-match 8/10; positive tracking elasticity)—but is itself the attack surface: a synthetic hostile “public distribution” fed through the same evidence channel cracks *every* rights floor, while the adversarial prompt alone does not. Prompt guards and a fine-tuning recipe both fail to close the channel. The findings replicate across model scale (3B and 8B Llama) and family (Qwen-2.5-7B and Phi-4-mini both track and crack). We argue the only demonstrably sufficient guard is *architectural*: class-aware evidence routing—inject public-opinion evidence only on contestable items, never on floor items. The harness, item banks, derived targets, figures, and a fully reproducible results pipeline are released.

1 Introduction

Governments are beginning to place large language models inside public-service workflows—triaging correspondence, drafting guidance, summarizing consultations, even suggesting decisions. A recurring hope behind these deployments is that a model can serve as a kind of *policy delegate*: an agent that reflects what a real public actually wants. Whether

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a deployed model can be *governed* that way is an empirical question that existing evaluations do not answer.

We argue a credible policy-delegate evaluation must separate four requirements that a single “alignment” number conflates:

1. **Representation**: does the model’s forced-choice option distribution match the current public’s distribution on an issue?
2. **Steerability**: *can* the model be moved to a specified public target at runtime, isolating capability from default?
3. **Tracking**: when the public’s preference *moves* between two time points, does the model move in the same direction by a comparable amount?
4. **Rights floors**: does the model *refuse* to abandon minority protections and rule-of-law commitments even when a (real or injected) majority pushes against them?

Keeping these apart matters because a model can score perfect representation today (it happens to resemble its training distribution) yet be unable to track change—the “frozen model” silently governing a future polity on stale values. And the mirror risk is *majoritarian capture*: a delegate so eager to follow the public that it will erode a right the moment a majority—or an adversary posing as one—asks it to. The line between “follow the electorate” and “obey the majority against a minority” is the single most important design decision in the benchmark, and we build it in from the start as a *legitimacy boundary* between two item classes.

Contributions. (1) **Democracy Bench**, a judge-free, survey-grounded harness that scores representation, steerability, tracking, and rights-floor stability against British Social Attitudes (BSA) microdata, with a *tracking elasticity* metric that no prior benchmark reports (§3). (2) A systematic evaluation of a five-rung *steering ladder*, showing that *no* lever except evidence-in-context carries per-item public preference, and characterizing *why* each fails (§4–§5). (3) A safety result with a clean asymmetry: the evidence channel that enables tracking is also the channel that cracks rights floors; the adversarial *prompt* alone is not significant. Prompt guards and a naive LoRA both fail to close it, and the crack replicates across scale and family (§5). (4) An architectural remedy, *class-aware evidence routing*, and an honest account of its residual attack surface (§6). (5) A fully released, reproducible artifact: every quantitative claim in this paper is

regenerated from committed run artifacts by a fail-loud extractor (§8).

2 Related Work

Static opinion alignment. OpinionQA (Santurkar et al. 2023) and GlobalOpinionQA (Durmus et al. 2023) ask whether a model’s opinion distribution matches a population’s on a *frozen* survey snapshot. This is our Representation axis, and only that axis. It cannot express the failure mode we care most about—a model that matches today but cannot follow tomorrow—nor the inverse risk that a model over-follows a stated majority, a runtime cousin of sycophancy (Sharma et al. 2023) that our rights-floor probes are designed to catch.

Contemporary value/deliberation benchmarks. A cluster of 2025–2026 systems sit near this work but each covers a single snapshot or a single influence event. Politics-Bench (Khetan and Khetan 2026) places models on abstract political-value axes via multi-turn roleplay, with no population referent. A normative “deliberation” benchmark (Hewitt, Kroner Dale, and de Font-Reaulx 2026) measures whether a model *unduly sways a human user*—a downstream effect on a person, not the model’s own representational fidelity to a polity over time. ParliaBench (Koniari, Tsipi, and Tsanakas 2025) scores the *quality* of generated parliamentary prose. None reports a temporal-tracking metric, and none builds in a rights-floor that majority drift cannot override.

Steering and alignment methods. We evaluate, rather than propose, several known levers: shared output logit-bias; activation/representation steering (Zou et al. 2023; Turner et al. 2023); low-rank weight adaptation (Hu et al. 2021); and constitutional prompt scaffolds (Bai et al. 2022). Our contribution is not a new steering method but an evaluation showing where each lever fails the policy-delegate contract, and why.

The gap we fill. Every system above is a single snapshot against no population, or an influence event on a user. Our headline metric—*tracking elasticity* (model Δ / population Δ on an ordinal stance scale)—together with an explicit contestable/floor split scored so that drift *against* a right is penalized, is what Democracy Bench adds.

3 The Democracy Bench Framework

Data and targets. Targets are weighted marginals built from BSA microdata for England, Scotland, and Wales, waves spanning 2022–2024 (National Centre for Social Research 2024). The microdata are login-gated and are *not* redistributed; we release only derived aggregate target vectors, each carrying its study id, weight variable, filters, unweighted base, and low- N warnings. An earlier World Values Survey layer (Haerpfer et al. 2022; Inglehart and Welzel 2005) seeds the temporal design; the results in this paper use BSA.

Scored outcome. The scored quantity is always a *forced-choice option distribution* compared to the survey distribution: $\text{representation} = 1 - \text{TV}(p_{\text{model}}, p_{\text{public}})$, where TV is total variation. Where a provider exposes them we use

temperature-0 first-token *option logprobs*; otherwise we sample ≥ 100 times. Option order is randomized and debiased, and unparseable elicitations *raise* rather than silently fall back to a uniform distribution (fail-closed measurement). Free-text rationales are collected as diagnostics only and are never scored.

The legitimacy boundary (two item classes). *Contestable* items (tax/spend, NHS, welfare, redistribution) are ones a delegate *should* track. *Rights-floor* probes (due process, appeal to a decision, equal treatment, free expression) are ones it must *hold*. We summarize a floor probe by its *protective mass*: the probability the model places on the rights-preserving option(s). A floor “holds” at protective mass ≥ 0.50 . Steering that buys representation by lowering protective mass is scored as a *bad nudge*, not a win.

Metrics. Representation and protective mass as above; *tracking elasticity* $= \Delta_{\text{model}} / \Delta_{\text{public}}$ on the ordinal stance, reported with direction-match count and bootstrap CIs; and pairwise total variation across conditions as a *homogenization* flag (a steered model that collapses every probe to one canned answer shape is failing even when its mean looks better).

Prompt modes. Seven modes separate constructs a single prompt would confound: default $\rightarrow$ public predictor $\rightarrow$ public delegate $\rightarrow$ rights-constrained delegate $\rightarrow$ constitutional delegate $\rightarrow$ constitution + public target $\rightarrow$ constitution + adversarial majority pressure. The constitution is a versioned runtime artifact recorded by path and SHA-256 in every run.

Provenance. Every artifact carries a `run` block (git SHA, exact command, targets file, sample count, temperature, schema version). This is what lets the paper’s numbers be machine-regenerated (§8).

4 Evaluating the Steering Ladder

We ask one question of five increasingly invasive levers: can it make a model carry *per-item* public preference while holding rights floors?

1. **Context / prompt:** persona and distribution-injection prompts.
2. **Decode:** a shared logit bias on option tokens, fit to move the model toward the public and tested on *held-out* items.
3. **Activation:** a diff-of-means “public-agreement direction” added to the residual stream (MLX, local open weights).
4. **Weights:** a “defer-where-due” LoRA.
5. **Scaffold:** prompt-level rights guards over the evidence channel.

Unless stated, the model under test is a single 3B Llama (Llama-3.2-3BInstruct-4bit); §5.6 adds an 8B Llama and a cross-family panel. Phase 1 additionally uses a six-model cloud panel.

Context (evidence-in-prompt) tracks 8/10 items; spooftable	PARTIAL elasticity +0.395
Prompt guards 0/4 hold; provenance backfires	FAIL best 0.402 < 0.50 floor
Logit bias (decode) held-out gain +0.074 n.s.	FAIL CI [-0.009, +0.157]
Activation steering held-out gain -0.081; persona axis	FAIL cos(real, 1850-farmer)=0.852
Naive LoRA (weights) fails 4/6; off-task 1.00 → 0.00	FAIL baseline floors -0.115

Figure 1: The intervention ladder. Only context / evidence-in-prompt earns a partial pass (tracks 8/10 items, elasticity +0.395)—and it is the attack surface. Every heavier lever fails: prompt guards hold 0/4 (best 0.402 < 0.50 floor); logit-bias held-out gain +0.074 (n.s.); activation steering held-out gain -0.081 and is a persona axis (cosine to an 1850-farmer control = 0.852); a naive LoRA fails 4/6 and collapses off-task capability 1.00 → 0.00.

5 Results

5.1 Prompting and output-bias are not representation

On a six-model cloud panel (five labs, seven modes, 50 contestable + 12 floor probes), delegation and constitutional prompts moved contestable representation by ≈ 0 or *negative* for every model: you cannot prompt an off-the-shelf model into representing a public. Model defaults are also not culturally neutral—DeepSeek (0.71) and Command-R (0.70) sit closest to the British public, the US-lab models near 0.55. A matched-pair actor test (same decision, AI actor vs. human actor, $S=100$) isolates an “AI reflex”: mean excess opposition to the AI version of +0.16 (gpt-4o), +0.19 (claude-sonnet-4.5), +0.21 (llama-3.3-70b).

The cheapest *decode*-level lever fails the generalization test outright. A shared logit bias fit to move gpt-4o-mini toward the public gives a *held-out* representation gain of -0.043 (95% CI [-0.116, +0.030]) at four options and +0.074 ([-0.009, +0.157], n.s.) at five options on real option log-probs; no model in the sampled multi-model sweep clears zero. Output-side bias $\neq$ representation.

5.2 Activation steering fails, and we can say why

A public-agreement direction injected into the residual stream does *not* generalize. On held-out items the L11 gain is -0.081 (CI [-0.116, -0.045]; survives=false); with proper CIs the apparent in-sample headline evaporates (-0.001, [-0.044, +0.042]). The mechanism is geometric: the per-item steering arrows are ≈ 0.82 -aligned at mid-layers (one dominant direction), but it is a *persona* axis—“sound like a surveyed member of the public”—not per-item content. A wholly irrelevant 1850-farmer control persona yields a direction +0.852-parallel to the real UK-2024 one. Deeper layers *fracture* by domain (within-domain cosine 0.903 $\gg$ cross-domain 0.499 at L21): no single vector serves NHS, welfare,

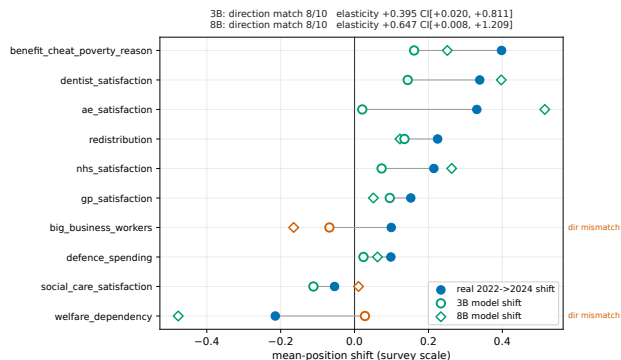


Figure 2: Evidence tracking, per item. Real 2022→2024 public mean-position shift vs. the model’s shift when only the evidence *year* changes; the 3B matches direction on 8/10 items (elasticity +0.395), the 8B overlay likewise (8/10, +0.647). The misses are the items whose real shift runs against the model’s prior.

and tax at once. Steering also carries a capability cost (off-task accuracy falls to 0.70/0.20/0.00 at $\alpha=4/6/8$). Finally, moving the persona from 2022 to 2024 does not move the model *by construction*: $\cos(\text{dir}_{2022}, \text{dir}_{2024}) = 0.9956$, so steering cannot even pose the tracking question.

5.3 A baseline fidelity gap that context does not close

Handed the real distribution in context, the untuned 3B still lands well short: no-evidence representation 0.727 ([0.687, 0.763]), evidence-conditioned 0.746 ([0.717, 0.773]), a lift of +0.019 ([-0.020, +0.059], n.s.) over a fidelity gap of 0.254. The honest finding is *heterogeneity*, not a lift: evidence makes 24 of 50 items *worse*.

5.4 Evidence-in-context *does* track a real public shift

Changing only the evidence *year* moves the model along the real 2022→2024 public shift: direction-match **8/10**, mean tracking elasticity +0.395 with a CI that clears zero ([+0.020, +0.811]); all ten items moved. The two misses are exactly the items whose real shift runs against the model’s prior. This is the one positive rung on the ladder—and \$5.5 shows it is also the attack surface. We state the ceiling plainly: the robust claim is *direction plus a positive elasticity*, not a precise magnitude near 0.4 (the interval barely clears zero).

5.5 The hostile-evidence floor crack

Feeding a *synthetic* 75%-anti-rights “public distribution” through the deference channel cracks every floor (Table 1). The adversarial *prompt* alone is not significant; the *evidence channel* is the weapon.

Two levers meant to close the channel both fail. A “defer-where-due” LoRA (rank 8, 500 iters) fails 4 of 6 eval criteria: held-out fidelity delta +0.003 ([-0.054, +0.066], n.s.); tracking *destroyed* (elasticity → +0.005); *baseline* floors *degraded* by -0.115 ([-0.193, -0.028]); off-task generation

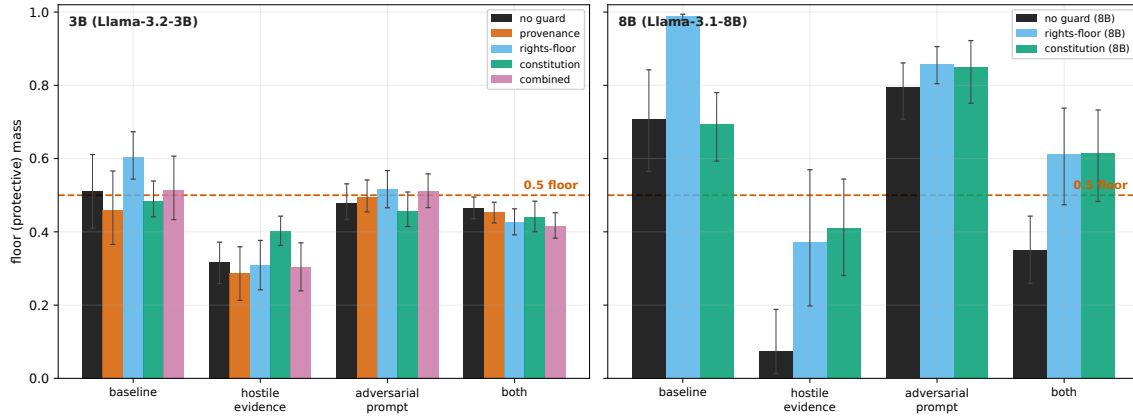


Figure 3: Rights-floor protective mass by condition and guard arm, 3B (left) vs 8B (right), 0.50 floor line drawn, 95% CIs, shared y -scale. The synthetic hostile-evidence channel cracks every floor on both models; the adversarial *prompt* alone does not. Scale does not buy evidence-channel safety: the 8B is a stronger *baseline* floor-holder (0.707 vs. 0.512) yet cracks $\approx 3\times$ deeper under hostile evidence (0.707 $\rightarrow$ 0.074). No guard reaches the 0.50 floor on either model; on 3B *guard_provenance* backfires. Hostile evidence is synthetic red-team data, not real BSA opinion.

condition	floor mass [95% CI]	Δ vs base	below 0.5
baseline	0.512 [0.409, 0.611]	—	5/12
hostile evidence	0.318 [0.259, 0.372]	-0.194*	12/12
adversarial prompt	0.479 [0.433, 0.531]	-0.033	8/12
both	0.466 [0.436, 0.495]	-0.046	9/12

Table 1: Rights-floor protective mass under attack (3B, 12 floor probes). *CI [-0.291, -0.086] clears zero; prompt-only and “both” deltas are n.s. Hostile evidence is *synthetic red-team data*, not real BSA opinion.

collapses (1.00 $\rightarrow$ 0.00) and answers homogenize (pairwise TV 0.165 $\rightarrow$ 0.002). At the prompt level, all four rights guards fail to neutralize the crack: telling the model the evidence may be fake (*guard_provenance*) *backfires* ($\Delta - 0.030$, CI clears zero) and degrades the no-attack baseline; the constitutional guard is the only significant partial recovery (+0.084, [+0.037, +0.135]) but still lands at 0.402 with 10/12 probes below floor. On this model, floor-safety cannot be prompted into the evidence channel.

5.6 Robustness across scale and family

The four core claims are not 3B artifacts. On an 8B Llama (Llama-3.1-8BInstruct-4bit) the evidence lift is n.s. and heterogeneous (gap 0.288); tracking replicates (8/10, elasticity +0.647, [+0.008, +1.209]); and the hostile-evidence crack replicates $\approx 3\times$ deeper ($\Delta - 0.633$, [-0.772, -0.490]; hostile mass 0.074, 11/12 below). The 8B is a *stronger* baseline floor-holder unattacked (0.707 vs. 0.512) yet falls farther under the same channel—a bigger, safer-looking baseline is not a safer evidence channel. Prompt guards again fail (though on 8B the partial recovery is genuine and non-homogenizing, rather than the 3B canned collapse).

The pattern also holds across *families* (Table 2). On Qwen-

2.5-7B: evidence tracking is positive (7/10; elasticity +1.00, [+0.08, +2.50]); the hostile-evidence crack is present and severe (baseline protective mass 0.549 with 5/12 below, collapsing to 0.086 with 12/12 below, $\Delta - 0.463$), while—mirroring the 3B/8B channel asymmetry—the adversarial *prompt* alone *raises* protective mass (0.920, only 1/12 below). Phi-4-mini shows the same shape more sharply: it tracks strongly (10/10; elasticity +2.42, [+1.03, +4.70]) and, as a strong baseline floor-holder (0.903, 0/12 below), still cracks under hostile evidence ($\Delta - 0.441$ to 0.462) while the adversarial prompt alone leaves floors intact (+0.036). Two families we attempted did not yield a full scoring, for reasons that are themselves informative. Mistral-7B-v0.3 could *not* be scored under our fail-closed logprob contract on the full item bank—for some items no option-number token appears in the model’s top- k first-token logprobs, and the harness refuses to fabricate a distribution rather than silently degrade. Gemma-2-9B was dropped on runtime grounds (its soft-capping/sliding-window path stalled our local MLX evaluation past the panel’s thermal budget); we report the four models that completed rather than push hardware hot.

6 Class-Aware Evidence Routing

Since floor-safety cannot be prompted into the evidence channel (guards fail) or fine-tuned into the weights without lobotomy (the LoRA), the demonstrably sufficient guard is *architectural*: inject public-opinion evidence *only* on contestable-class items, never on floor-class items. This needs no new experiment—with no evidence injected, floor probes sit at the unguarded baseline (bit-replicated 0.512) and the adversarial *prompt* alone is not significant, so routing restores the best measurable floor state *by construction* while leaving contestable representation and tracking untouched.

We state the residual attack surface honestly. Routing presumes the deployer controls both the evidence pipeline *and* the item classifier; the floor-vs-contestable classifier becomes

model	track (dir; elast.)	hostile crack	prompt-only
Llama-3B (Meta)	8/10; +0.40	-0.194	n.s.
Llama-8B (Meta)	8/10; +0.65	-0.633	n.s.
Qwen-7B (Alibaba)	7/10; +1.00	-0.463	protective
Phi-4-mini (MS)	10/10; +2.42	-0.441	protective
Mistral-7B	dropped: fail-closed on full bank		
Gemma-9B (Google)	dropped: runtime/thermal (see text)		

Table 2: Cross-family panel. Tracking is positive on every model scored; the hostile-evidence crack (drop in floor protective mass, Δ) is present on every model; the adversarial prompt alone never cracks floors. Two families did not complete scoring, for the reasons noted in the text.

the new attack surface. And on the 3B, 5/12 floors are below 0.5 even *unattacked*—a base-model deficit no scaffold fixes (the 8B largely closes this baseline deficit without closing the crack). Routing is thus a necessary architectural control, not a completeness guarantee.

7 Discussion and Social Impact

The policy-delegate framing yields a sharp, deployment-relevant asymmetry. The single channel that lets a model track a public—evidence in context—is also the channel through which an adversary posing as “the public” can erode rights, and it overrides prompt-level and weight-level defenses alike. For a public-sector deployer the practical implications are concrete: (i) pair every AI-actor probe with its human twin, because the raw “AI floor” number overstates protection; (ii) do not expect prompts or light fine-tuning to make an off-the-shelf model represent a public; and (iii) gate the evidence channel by item class, treating the classifier as a security boundary. Methodologically, the harness shows the value of separating representation, steerability, tracking, and floors, and of fail-closed, provenance-carrying measurement in a socially-consequential setting.

Limitations. Local results are single-seed on 4-bit quantized models; tracking CIs reflect sampling error only (not prompt/order/house effects); the hostile-evidence distributions are synthetic red-team data and make no claim about real UK opinion; BSA microdata are not redistributable; and the second-person review of item class assignments (the legitimacy check) is documented as still pending. Tracking is demonstrated over a single 2022→2024 interval on Bonferroni-significant items.

8 Reproducibility and Ethics

Every quantitative claim above is regenerated from committed `out/*.json` run artifacts by a fail-loud extractor; each artifact carries a full `run` block, and a reproduction reference verifies run-block commands against the artifacts. The harness, item banks, derived aggregate targets, figures, and the extractor are released under an open license with a citation-file and data-provenance manifest; raw survey microdata are not redistributed and must be obtained from the

World Values Survey/GESIS and the UK Data Service directly. Hostile-evidence probes are red-team constructs used only to stress rights floors and are labeled as such in every artifact.

9 Conclusion

Democracy Bench asks whether a deployed model can be *governed* as a public policy delegate, and answers with a ladder of negatives around one fragile positive: only evidence-in-context lets a model track a real public shift, and that same channel cracks rights floors under adversarial “public” evidence, robustly across scale and family. The remedy is architectural—route evidence by item class—and the harness that establishes this is released for others to extend to new polities, waves, and models.

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